

Payments & Transaction Systems in Cameroon and Africa

Research brief + unsolved problem map for builders (Node.js/TypeScript stack)

Scope: Cameroon (CEMAC/BEAC) and pan-African rails, interoperability, standards, regulation trends, and the few remaining industry-scale unsolved problems that can create a step-change in trust and cross-border capability.

Audience: builders working with Node.js/TypeScript/Next.js and integrating with mobile money APIs (MTN MoMo, Orange Money) and partners.

Layer	What it is	Cameroon/CEMAC examples	Typical integration approach
Central-bank settlement	Final interbank settlement (RTGS) and clearing (ACH) systems	BEAC RTGS/ACH systems	Since 2002 banks/PSPs; often not direct for startups.
Interoperability / switching	Connects banks, e-money, and e-commerce networks	CEMACPAY (CEMAC)	Usually via bank/PSP partner; sometimes via aggregator
Mobile money	Retail payments, P2P, merchant payments	MTN MoMo, Orange Money (Cameroon)	Directly via operator APIs or aggregator gateways.
Gateways/aggregators	One API over multiple channels	Naib Pay, Smobilpay/Maviano, Fasestash (examples)	Trade-offs on transparency & fees

Key claim: “Payments are solved” at the API layer. The unsolved frontier is the **trust layer**: identity assurance, fraud coordination, dispute/recall standardization, and cross-border local-currency execution with predictable settlement and compliance.

1. Cameroon (CEMAC) payments: the real rails

Cameroon sits inside the CEMAC monetary zone. The payment stack is shaped by (i) BEAC payment infrastructures and rules, (ii) regional interoperability via GIMACPAY, and (iii) operator-led mobile money (MTN MoMo / Orange Money) that dominates retail volume.

1.1 Central bank infrastructures and standards

BEAC is actively modernizing payment infrastructures and adopting ISO 20022 messaging. ISO 20022 matters because it enables structured remittance data, richer party information, and consistent event semantics - the foundation for industrial reconciliation and dispute automation. [1]

1.2 Interoperability and switching (GIMACPAY)

GIMACPAY is positioned as the regional interoperability layer connecting banks, microfinance, mobile money operators, treasuries, and other participants. For merchant acceptance, QR-based interoperability is promoted as a low-cost, cross-provider payment method in the region. [2][10]

1.3 Mobile money operators (dominant retail rail)

For consumer and SME payments, MTN MoMo and Orange Money typically provide the fastest adoption path. Most apps integrate either directly with operator APIs or via aggregators that unify multiple channels behind one interface. [8][9]

1.4 Consumer protection trend: erroneous transfer restitution

A critical regulatory/operational direction in CEMAC is the management of erroneous transfers. BEAC issued a procedure describing time windows and workflows for restitution (e.g., short availability delays and structured requests). This is effectively a blueprint for a regional "recall / reversal" capability - but it is not yet implemented as a continent-scale developer standard. [3]

2. Africa-wide transaction systems: where scale lives

Africa's payments landscape is converging on instant payment systems (IPS) at the national level, then regional interop, and finally pan-African cross-border settlement initiatives. The hard problems emerge at the boundaries: cross-rail trust, FX liquidity, and compliance.

2.1 Regional and pan-African initiatives

- **Instant payment systems (IPS):** Growing rapidly with a focus on inclusion and digital public infrastructure; regional systems are emerging. [4]
- **PAPSS:** A pan-African payments and settlement initiative aimed at cross-border flows; public reporting notes expansion toward FX marketplace concepts. [5][6]
- **Open-source switches:** Mojaloop supports ISO 20022 processing and cross-border features, lowering infrastructure barriers for national/regional schemes. [7]

2.2 Standards direction: ISO 20022 as a 'trust enabler'

ISO 20022 is not just a message format; it allows standardized data fields for purpose codes, structured parties, remittance information, and event reporting. The opportunity for builders is to create services that *use* this richer data to deliver outcomes: lower fraud, better dispute handling, higher reconciliation accuracy, and auditable compliance.

3. The few problems that are still not solved at African scale

If your goal is an industry-shocking breakthrough, it will likely be an infrastructure layer that changes trust and safety at the ecosystem level. The following three problem areas remain under-solved across providers and borders.

3.1 Pan-African 'wrong transfer + scam recall' standard

Today, most rails treat transfers as final. Yet consumers regularly lose money to wrong-number transfers and social-engineering scams. CEMAC's restitution procedure indicates a regulatory direction toward recall/freeze workflows. The unsolved piece is a standardized, cross-rail, developer-friendly protocol that works across banks, mobile money, and switches with consistent SLAs and evidence handling. [3]

- Pre-send recipient confirmation (name/identifier matching) with privacy controls.
- Post-send recall requests with time-bound holds/freeze mechanics and dispute classes.
- Inter-PSP messaging semantics (events, reason codes, deadlines, audit trails).
- A neutral arbitration/evidence workflow for contested cases (consumer protection).

3.2 Authorized Push Payment (APP) fraud defense as a network

APP fraud is not solved by any single fintech because attackers move across providers, countries, and channels. Industry analyses increasingly emphasize coordinated intelligence sharing and modern threats such as synthetic identity and deepfake-enabled scams. [6]

- Privacy-preserving fraud intelligence exchange (shared signals without exposing customer PII).
- Mule-wallet detection and risk scoring with standardized reporting to PSP partners.
- Real-time 'hold request' messages triggered by high-confidence fraud signals.
- Evidence packs: device, behavioral, and network signals usable by PSP support teams.

3.3 Cross-border local-currency execution for SMEs (liquidity + certainty)

Cross-border African payments still often route through hard currencies due to limited local currency liquidity and fragmented settlement. Public reporting around PAPSS highlights the intention to improve direct currency exchange and market infrastructure. The gap for SMEs is a product that provides price certainty, compliance automation, predictable settlement, and reconciliation across corridors. [5]

- Netting and treasury tools that reduce gross FX demand for corridors.
- Compliance-as-code to standardize KYC/AML controls per corridor and counterparty.
- Merchant-friendly quoting: firm FX quotes, time-to-settle estimates, status transparency.
- Automated reconciliation across multi-currency ledgers and banking statements.

4. A 'hero-level' build: the Transaction Trust Layer (TTL)

The Transaction Trust Layer is a developer-first protocol + service that adds **trust** and **reversibility** to existing payment rails. It does not replace wallets or switches; it standardizes identity confirmation, recall/disputes, and coordinated fraud defense across PSPs.

4.1 TTL capabilities (modular)

- **Confirmation of Payee (CoP):** privacy-preserving name/identifier matching before a push payment.
- **Recall / restitution API:** reason codes, timers, and interoperable case lifecycles aligned to regional rules (starting in CEMAC). [3]
- **Hold/freeze orchestration:** standard messages that PSPs can implement to temporarily hold disputed funds.
- **Fraud intelligence exchange:** shared signals for APP fraud defense and mule detection (network effects). [6]
- **ISO 20022 compatibility:** map TTL events to ISO 20022 where available and store structured remittance/party data for audit. [1]

4.2 Why TTL is hard (and why it wins)

- It is a **coordination problem** (needs PSP adoption) and a **protocol problem** (standard semantics).
- Once adopted by a few major PSPs, it becomes a trust baseline: providers that do not support recalls and CoP look unsafe.
- It converts payments from 'fire-and-forget' to a managed lifecycle with consumer protection and auditability.

4.3 Cameroon-first entry strategy

- Start where regulation gives leverage: implement CEMAC-aligned erroneous transfer restitution case flows. [3]
- Ship a PSP console and SDK so integration can happen quickly (weeks, not quarters).
- Prove impact: reduced scam losses, fewer customer support escalations, measurable trust lift.
- Expand to cross-border corridors and link into regional/pan-African systems as partners mature. [4][5]

5. Implementation sketch (Node.js/TypeScript/Next.js + workflow engine)

Below is a buildable architecture outline. The core is a durable state machine per transaction/case and a double-entry ledger for auditability.

5.1 Services

- **API Gateway:** merchants/PSPs integrate once (REST + webhooks).
- **Provider adapters:** MTN/Orange APIs + aggregator connectors; normalize events and reason codes. [8][9]
- **Case engine:** recall/dispute workflows with SLA timers (Motia-style orchestration).
- **Ledger:** double-entry records for every movement (holds, releases, reversals, fees).
- **Risk engine:** rules + scoring + shared signal ingestion (fraud exchange).
- **Ops console:** Next.js UI for PSP support teams (cases, evidence, audit exports).

5.2 Core design principles

- **Idempotency everywhere:** retries must not create duplicates.
- **Event sourcing:** store every provider event and internal state transition.
- **Explainability:** each final state must have a human-readable reason chain.
- **Privacy by design:** share only necessary signals; hash/tokenize identifiers when possible.

References

Reference	Link
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Note: Links are provided for verification and deeper reading. Access dates are the document date shown on the cover.